

# iverilog vs Verilator

Before you compile anything, pick the right engine for the job

# Match the scenario

- Use Icarus when the lesson is hash delays, initial blocks
- Use Verilator when the job is a Makefile regression farm
- The browser lab presents matched scenarios: same design idea, different tool fit
- Read the scenario before you click a answer

# Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

iverilog vs Verilator chooser

Document-aid literacy: pick **iverilog** or **Verilator** from a use-case matrix (not a real simulator run). Starter: long regression → Verilator — MATCHED.

Starter example: use case long\_regression → tool Verilator — MATCHED.

Load starter example

Challenges 0 / 22 cleared

Quiz: iverilog: Prefer iverilog when you need...

☐ event-driven semantics, #delay / races, or easy course vvp+VCD

☐ only the fastest huge farm regression

☐ a C++ SystemC host only

☐ place-and-route

Show hint

Check

Next

Idle

1

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5

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Core ideas

iverilog

Event-driven · delays · course VCD.

Verilator

Cycle speed · C++ host · big regs.

use case

Job first — then pick the tool.

MATCHED

Tool fits the matrix suggestion.

Lab

SCENARIO PACK

speed → Verilator

Load preset

Choose

Scan choice

Demo mismatch

Explain

Reset

Lab starter

# Real Verilator practice

- In Track A, open this module's EXAMPLES prompts and the legacy course tree
- Run one tiny design with Icarus for event-style sanity
- You do not need a perfect port

# Pitfalls to watch

- Do not reach for Verilator when the lesson requires hash delays and timing checks Icarus
- Do not reach for Icarus when you need fast cycle regressions at scale
- Do not assume “simulator is simulator”, paradigm mismatch wastes afternoons
- And do not treat the browser lab as proof your install works; that is Track A’s job

# Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish matched scenarios with ready status
- In Track A, name one job for each tool from your own work
- When you are ready, take the short quiz, then continue to the simulation pipeline